

Introduction to Machine Learning

By

Prof(Dr) Premanand P Ghadekar

Outline

- ❖ Introduction to ML
- ❖ Syllabus
- ❖ Data Science Process Flow chart
- ❖ Best Practices for Data Science Projects
- ❖ Data Science based Project ideas
- ❖ Impact of Artificial Intelligence on Industries
- ❖ Artificial Intelligence Engineers Skill Set
- ❖ Job Roles & Opportunities for Students
- ❖ Research Opportunities etc.



Artificial Intelligence

18

TIMES TECHIES

THE TIMES OF INDIA
WEDNESDAY, JANUARY 1, 2025

With 2025 upon us we asked senior tech execs what they thought were some important tech trends to watch out for. Most of them, predictably, spoke of the growing dominance of AI in every aspect of our lives. From custom AI agents and AI PCs to more linguistically diverse platforms, AI is set to dominate the tech scene for another year, at least.

AI WILL DOMINATE 2025

AI on the edge is coming

One of the most exciting areas of progress in the field of artificial intelligence is edge AI, where devices and IoT gateways can handle AI tasks right where they are—locally. With hardware like NPUs becoming a standard feature in devices, real-time applications such as language translation and advanced vision processing are becoming more efficient and accessible. This evolution not only improves performance but also enables businesses of all sizes to leverage AI effectively. Companies like us are driving advancements in edge AI by equipping developers with tools and platforms to address local and complex challenges. In India, the pace of digital transformation is picking up quickly, highlighting the importance of solving AI in a timely and effective way. With its unique demographics and dynamic markets, India has the potential to become a global leader in AI innovation.



Rajeev Kumar
CEO and
Founder,
SaaS

Human language will become less critical to Agentic AI proliferation

As developers continue to train AI models on increasingly vast amounts of content, the progress of human language. This creates two significant challenges that AI solution providers must address in the coming year. Firstly, I foresee that content creators and model developers will need to find ways to keep up with the rapid pace of AI evolution. In 2025, large AI developers and data providers are seeing the pushback from artists, musicians, and consumers as they use their



and more human-created content to train their models. The demand for AI model training data is enormous, with one estimate suggesting that by 2025, developers will require 100 terabytes of data to train their models. As AI's demand for human-generated data grows, the value of this data and the influence of those who control it will be significantly amplified. I believe this issue along with other language-related constraints will push AI models to evolve beyond human language by 2025.

Pooja Kohli | CEO, Automation Anywhere

UPI 3.0 and AI for Bharat

UPI 3.0 is poised to revolutionise digital transactions in India by introducing features that enhance user convenience and financial inclusion. Key innovations include conversational payments, enabling users to conduct transactions through voice interactions with AI-powered systems, initially supporting Hindi and English. This advancement aims to make digital payments more intuitive and accessible to a broader user base. Additionally, the integration of credit lines allows users to access pre-sanctioned credit directly through their UPI apps, providing greater financial flexibility. This feature facilitates seamless borrowing and spending without the need for separate credit applications. The rise of Gen AI models tailored for India's diverse linguistic landscape is also set to democratise digital access. Initiatives like Sarvaat AI offer low-cost, high-quality models across 10 major Indian languages alongside English. Similarly, the Indian government's Bhaskari project aims to develop a National Public Digital Platform for languages, facilitating seamless translation and interpretation in various Indian languages. These advancements are expected to bridge linguistic divides, enabling more Indians to leverage AI in their native languages.

Rohan Sachdev | Founder and National Lead, Consulting Services, by India

The Year of Synthetic Data and Quantum Innovation

A transformative year defined by the interconnected trends that will reshape industries and amplify innovation. The first is Models as Products. The shift toward deploying pre-trained, specialised models will define AI adoption. Beyond generative AI, organisations will rely on deterministic models for fraud detection, optimisation, and more, allowing for scalable and actionable insights. The second trend is the Synthetic data revolution. Synthetic data sets are becoming indispensable in addressing privacy concerns and enhancing AI performance. Industries like healthcare and finance will see up to 50% improvements in model accuracy, thanks to synthetic data's ability to simulate rare events and refine predictive capabilities. The third trend relates to AI governance. As AI decision-making permeates critical areas like lending and making predictions, organisations will prioritise oversight, platform compliance, and ethical considerations. Equally important will be AI cloud deployment. AI at scale, managing cloud expenses will take centre stage. Organisations will explore innovative strategies to balance performance with cost efficiency, ensuring sustainable AI operations. On top of that, Quantum computing will unlock new frontiers in AI and data analysis. Initiatives will begin leveraging quantum capabilities to tackle complex problems, setting the stage for breakthroughs in optimisation and computational speed.

Bryan Harris | Chief Technology Officer, SaaS



Small models will dominate

2024 has been a transformative year for AI innovation and this momentum will continue in 2025. India will continue to demonstrate AI, as we develop cost-effective Small Language Models (SLMs) and multilingual models to cater to the linguistic diversity of the country. SLMs will see increased adoption in 2025 by enterprises across all major sectors, which will benefit from its increased efficiency, low compute and personalisation nature. Emerging technology capabilities like multi-modal input and output, deeper reasoning and processing rich content will go mainstream and pave the way for an Agentic world where AI agents can act on our behalf across our life, work and organisations.

Rajeev Kumar | CEO and Founder, SaaS and Automation Center

AI PCs will continue to evolve

Believe me, we are truly at the intersection of multiple breakthroughs in AI. Services like AI PCs and the rapid expansion of High-performance computing (HPC) in the cloud and edge. AI PCs represent a transformative shift from incremental upgrades to devices that can process data locally, reducing latency and enabling real-time processing. These innovations deliver faster response times, improved data security, and the ability to handle complex tasks offline, reshaping everyday workflows. For instance, imagine a sales rep visiting a client. AI PCs can streamline tasks, extend battery life, and even provide real-time translations. All of this without connecting to the cloud. Meanwhile, AI workloads and machine learning will grow exponentially. The expansion of HPC with AI accelerators will enable enterprises to manage increasing complexity, powering AI simulations, and these trends are all part of a broader shift toward making cutting-edge tech more accessible and seamlessly integrated into AI-led devices, redefining how we connect, and innovate.



Sandhya Arora
CEO, Wipro

Expect to see custom AI solutions

As we look ahead to 2025, AI isn't just an enabler; it's becoming the backbone of how industries innovate, scale, and solve their most complex challenges. Custom AI solutions are expected to see widespread adoption, moving from being a niche offering to becoming truly ubiquitous. With businesses agents capable of making decisions and extending entire ecosystems. Custom AI in particular is emerging as a transformative force in 2025, allowing organisations to design AI solutions tailored to their specific needs, by leveraging domain-specific data and advanced model customisation. Custom AI empowers organisations to address unique challenges, from personalised customer experiences to industry-specific innovations. Industry analysts underscore the growing demand for differentiated AI solutions as enterprises seek to differentiate themselves in competitive markets. Custom AI empowers organisations to address unique challenges, from personalised customer experiences to industry-specific innovations. Industry analysts underscore the growing demand for differentiated AI solutions as enterprises seek to differentiate themselves in competitive markets.

Sandhya Arora | CEO, Wipro



Extended reality will finally be here

The usage of extended reality will increase. In this evolving landscape, multimodal AI will seamlessly integrate into real-world human interactions and navigation. Technology will become more intuitive and responsive to human needs, including security and privacy. Offering enterprises new opportunities to elevate customer experiences and empower employees. For example, spatial computing will augment customer service, super experiences, travel experiences, patient-provider interactions, and remote work. XR and AI-powered solutions for inspection and predictive maintenance will impact safety and environmental protection in addition to operational efficiency. We will also witness an increase in autonomous machines with over-the-air (OTA) updates. Autonomous industrial robots will proliferate, and Software Defined Medical Devices will evolve towards autonomous preventive maintenance and self-healing with minimal human intervention and downtime.

Sandhya Arora | CEO, Wipro

Datacentres will mushroom everywhere

As we look ahead to 2025, we at Intel believe that expanding data centre capacity is not just critical, it is non-negotiable to meet the demands of digital India. India contributes to approximately 20 percent of the world's data, yet most of it is stored and processed outside the country's borders. We would not only be contributing to security, compliance, and good governance, but this would also significantly reduce the latency of training localised AI models using this data. This would also yield an overall cost-effective model, as transferring large volumes of data to overseas data centres will incur hefty recurring costs due to sub-optimisations. With the growth in localised data centres, we can expect to come down further from economies of scale as more data centres. The tech ecosystem needs to focus on reimagining the security aspects in addition to addressing some of the environmental challenges like water and power. Investing in local centres—also to build highways or other infrastructure, is critical to India's future as the nation aims to be an AI powerhouse.

Sandhya Arora | CEO, Wipro



AI specific servers will be in demand

2024 has seen a stronger demand for AI servers, fuelled by the increased adoption of AI and generative AI models. As these servers consume significantly more power and generate more heat than standard servers, we can expect to see increased demand for AI-specific infrastructure in the coming year. These servers are designed to handle the unique demands of AI workloads, with optimised storage and memory to enable high-speed server-to-server communication, scalable storage systems, and efficient power management. The challenge for businesses will be on how they manage the high implementation and maintenance costs of this infrastructure. Employing a hybrid cloud strategy, combining on-premises infrastructure with cloud-based AI services, can help reduce these costs while providing scalability. In 2025, we will also see increased scrutiny on businesses to act in an environmentally sustainable manner, with some governments introducing legislation imposing tighter restrictions on the disclosure of climate risks. For data centres, the challenge will be balancing the need for innovation and scalability with its carbon footprint.

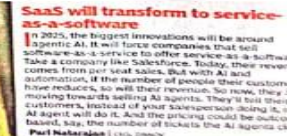
Amir Luthfi | Managing Director, India, United World



Software and Hardware as a Service

In 2025, technology is changing at a pace never seen before, and SaaS (Software and Hardware as a Service) is poised to revolutionise the way enterprises adopt and use technology. Emerging Generative AI platforms showcase the groundbreaking potential of this approach, enabling businesses to create, deploy, and scale solutions in a matter of minutes. What used to require years to innovate now becomes a weekly occurrence, so the pace of new, creative, disruptive technologies is unprecedented. Autonomous solutions for software, powered by AI, IoT, blockchain, and cloud reality, will be the future—shaping tomorrow's service, which goes beyond the very concept of SaaS itself. This trend brings the biggest challenge for the old players in the software space because it makes possible deployment of fully integrated solutions almost overnight. SaaS is the next wave of innovation that can revolutionise and unlock the possibilities of any industry across the globe.

Prakash Sornali | Founder, CEO & CTO, CDS Software Solution



SaaS will transform to service-as-a-software

In 2025, the biggest innovations will be around Agentic AI. It will force companies that sell software-as-a-service to offer service-as-a-software. Take a company like Salesforce. Today, their revenue comes from per-seat sales. But with AI and automation, if the number of people their customers have reduces, so will their revenue. So now, they are moving towards selling AI agents. They're not their customers, instead of your salesperson doing it, it's the AI agent who'll do it. And the pricing could be outcome based, say, the number of tickets that AI agents solve.

Priyanka Natarajan | CEO, Oracle

Terminology

Machine Learning, Deep Learning, Data Science, Data Mining, Data Analysis, Statistical Learning, Knowledge Discovery in Databases, Pattern Discovery, AI.



Smile, we are 'DATAFIED'!

- ❖ Wherever we go, we are “datafied”.
- ❖ Smartphones are tracking our locations.
- ❖ We leave a data trail in our web browsing.
- ❖ Interaction in social networks.
- ❖ Privacy is an important issue.

Data everywhere!

1. **Google:** processes 24 peta bytes of data per day.
2. **Facebook:** 10 million photos uploaded every hour.
3. **Youtube:** 1 hour of video uploaded every second.
4. **Twitter:** 400 million tweets per day.
5. **Astronomy:** Satellite data is in hundreds of PB.
6. . . .
7. **“By 2026 the digital universe will reach 60+ zettabytes...”**

Giga, Tera, Peta, Exza, Zeta & Yotta

Data types

Data comes in different sizes and also flavors (types):

- ☒ **Texts**
- ☒ **Numbers**
- ☒ **Clickstreams**
- ☒ **Graphs**
- ☒ **Tables**
- ☒ **Images**
- ☒ **Transactions**
- ☒ **Videos**
- ☒ **Some or all of the above!**

Applications of ML

- We all use it on a daily basis. Examples:



Machine Learning

- ❖ Spam filtering
- ❖ Credit card fraud detection
- ❖ Digit recognition on checks, zip codes
- ❖ Detecting faces in images
- ❖ MRI image analysis
- ❖ Recommendation system
- ❖ Search engines
- ❖ Handwriting recognition
- ❖ Scene classification
- ❖ etc...

What is Machine Learning?

“How do we create computer programs that improve with experience?”

- Tom Mitchell
 - http://videolectures.net/mlas06_mitchell_itm/
-
- ❖ **Machine Learning** is the Science of making computers learn and act like humans by feeding data and information **without being explicitly programmed.**
 - ❖ A branch of **artificial intelligence**, concerned with the design and development of algorithms that allow computers to evolve behaviors based on empirical data.

Why Machine Learning ?



Identifying The Factors/ Opportunities

Which Can Be Further Used For Decision Making And Value Creation.

Why “Learn”?

- ❖ **Machine Learning** is programming computers to optimize a performance criterion using example data or past experience.
- ❖ **Learning is used when:**
 - Human expertise does not exist (Navigating on Mars),
 - Humans are unable to explain their expertise (Speech Recognition)
 - Solution changes in time (Routing on a Computer Network)
 - Solution needs to be adapted to particular cases (user Biometrics).

Machine Learning

Prerequisite- Linear Algebra, Statistics, Probability, Calculus, and Programming Languages

Unit I -Introduction of Machine Learning

Unit II - Supervised Learning

Unit III - Statistical Learning

Unit IV- Naive Bayes and Support Vector Machine

Unit V- Decision Trees and Ensemble Learning

Unit VI- Unsupervised and Reinforcement learning

Machine Learning & Deep Learning Books

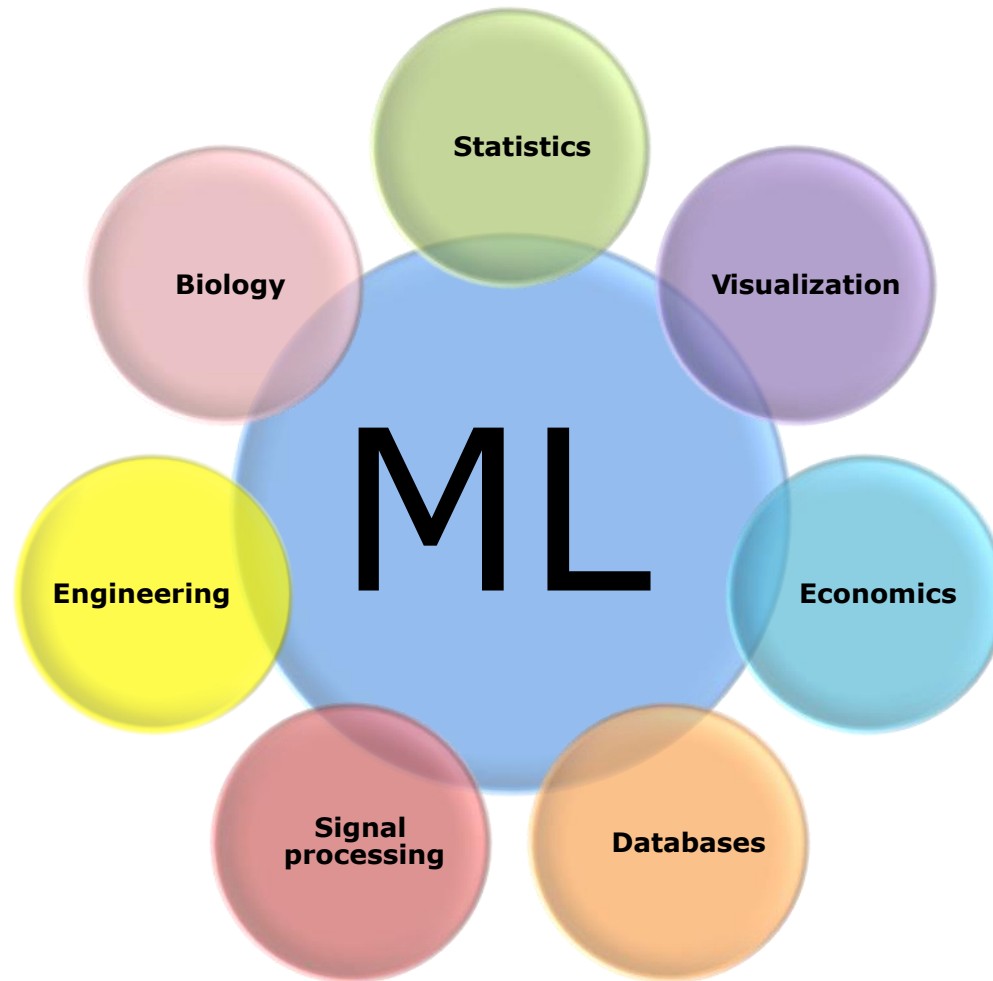
Text Books:

1. T. Mitchell, "Machine Learning", McGraw-Hill, 1997.
2. Anup Kumar Srivastava, Soft Computing, Alpha Science International limited. 2009.

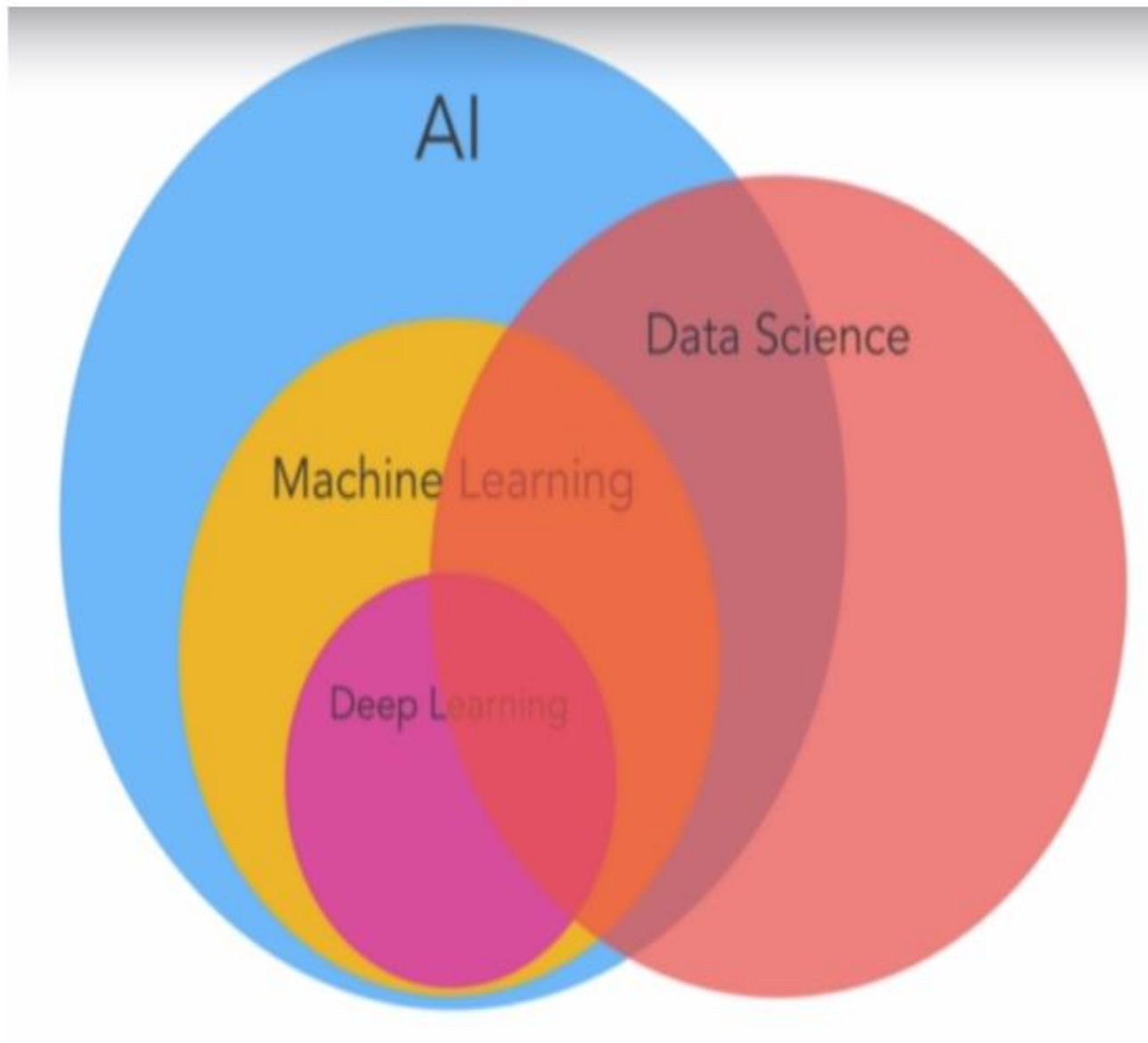
Reference Books:

1. Ethem Alpaydin, "Introduction to Machine Learning", MIT press, 2004.
2. Jacek M. Zurada, "Introduction to Artificial neural System", JAICO publishing house, 2002.

Interdisciplinary field



AI, ML and DS

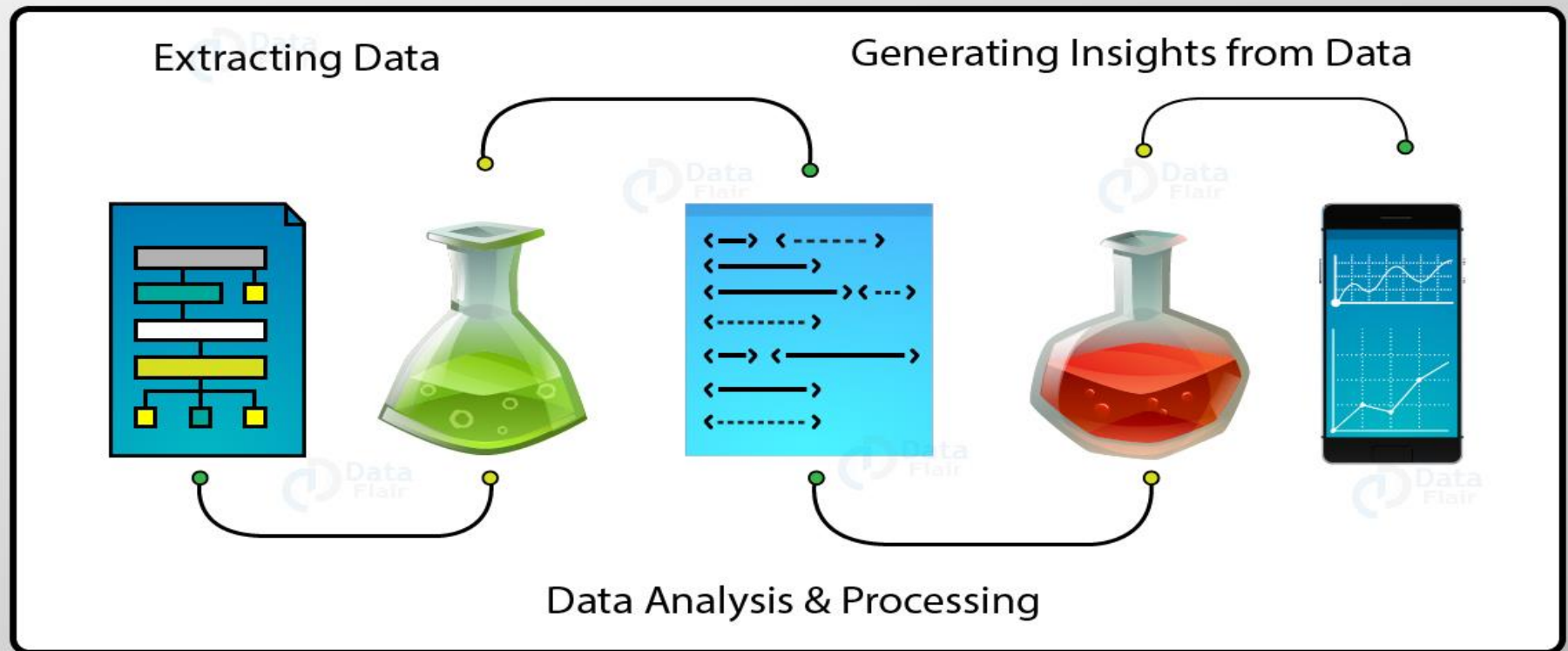


Why Data Science

- ❖ In the last two years alone, **90% of the world's data** has been created.
- ❖ **2.5 Quintillion bytes of data** are produced by humans every day.
- ❖ **Huge Amount of data** and **High Performance Computing** facilities are available.
- ❖ **Data Science** is the field of study that combines domain expertise, programming skills, and knowledge of mathematics and statistics **to extract meaningful insights from Data.**



Purpose of Data Science



Introduction

- ❖ **Data Science** is rapidly growing to occupy all the industries of the world.
- ❖ The **future is Automation** with tremendous scope for Data Science
- ❖ **Data creates magic.**
- ❖ **Industries** need data to help them make **Careful Decisions.**
- ❖ Data Science churns raw data into meaningful insights. Therefore, industries need data science.

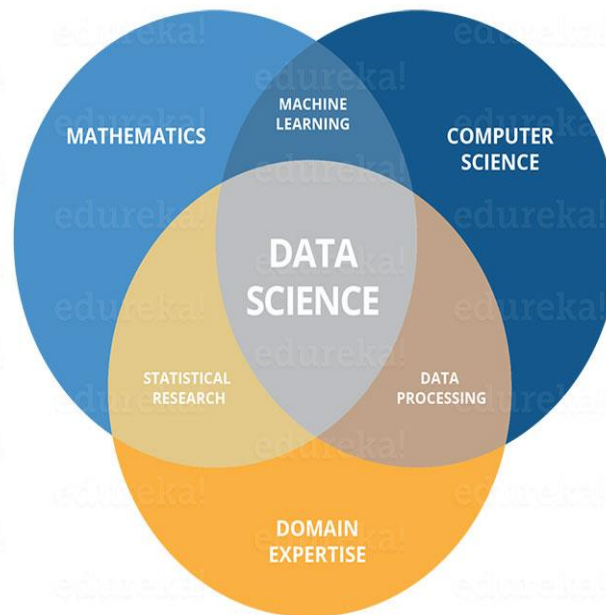


Importance of Data Science in Business



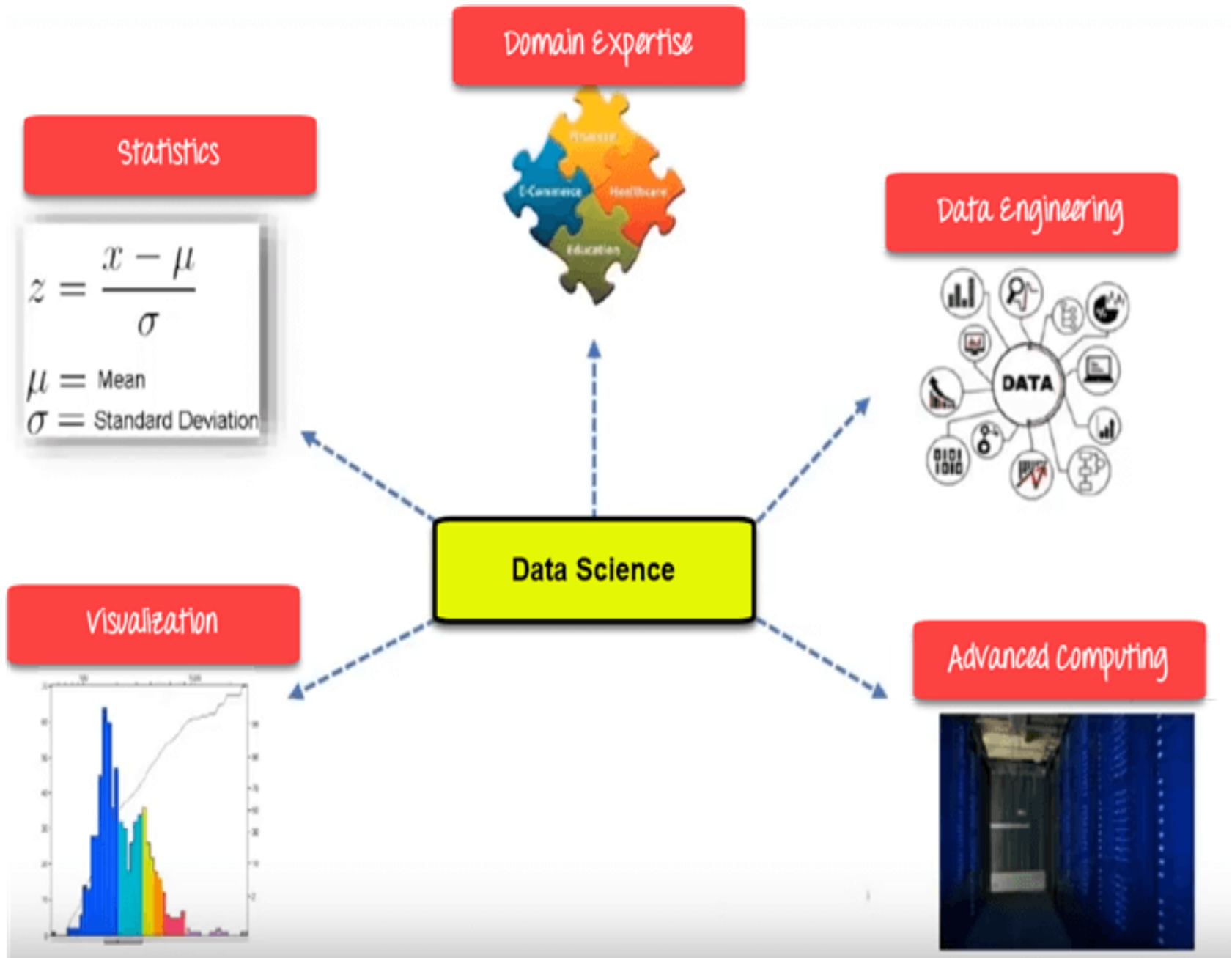
What is Data Science

- **Data Science** is the science which uses **Computer science, Statistics, Visualization and Machine Learning Principles** to **collect, clean, integrate, analyze, visualize, and to interact with data** to **create data products**.

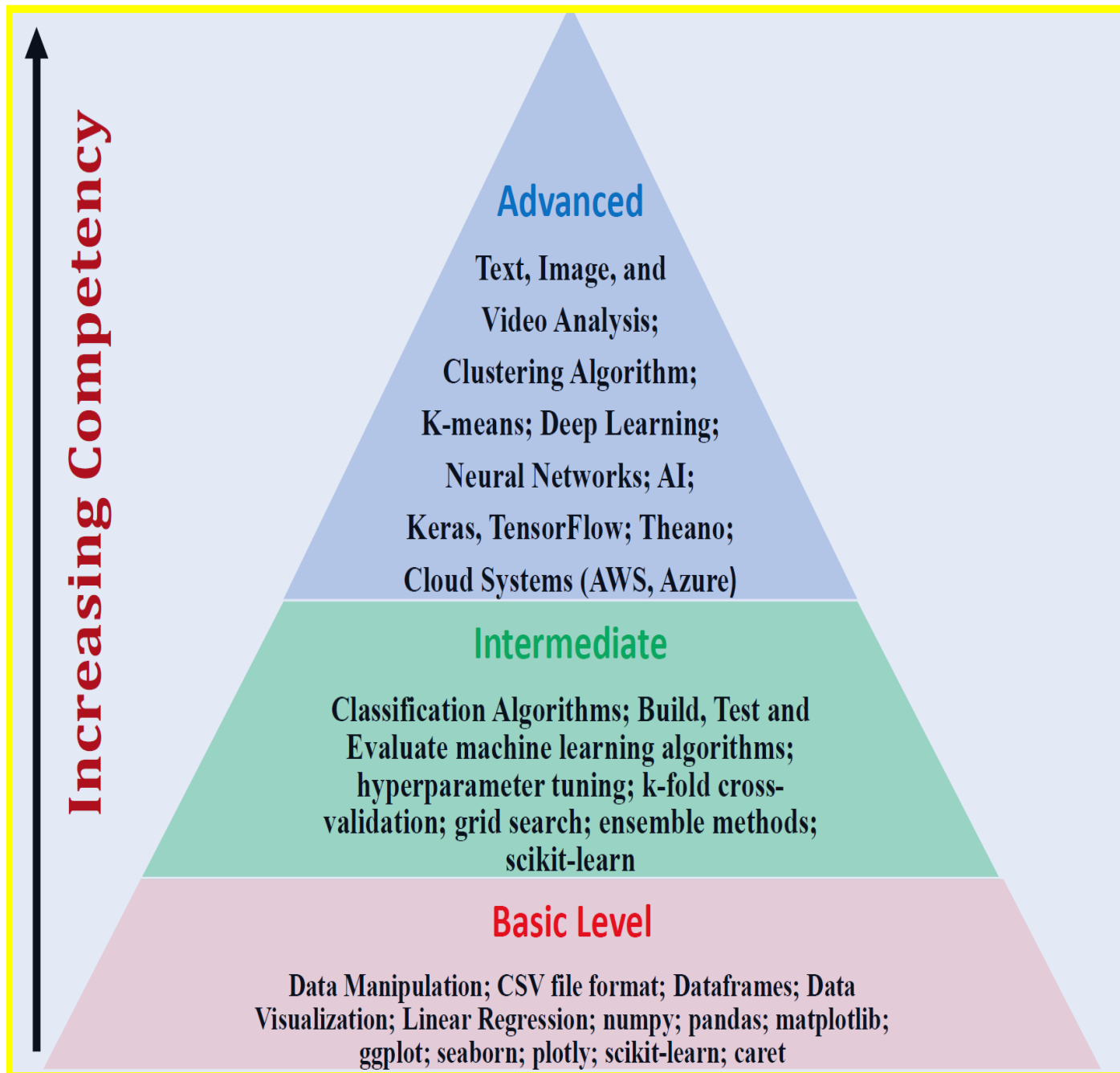


Goal of Data Science - Turn Data into Data Products

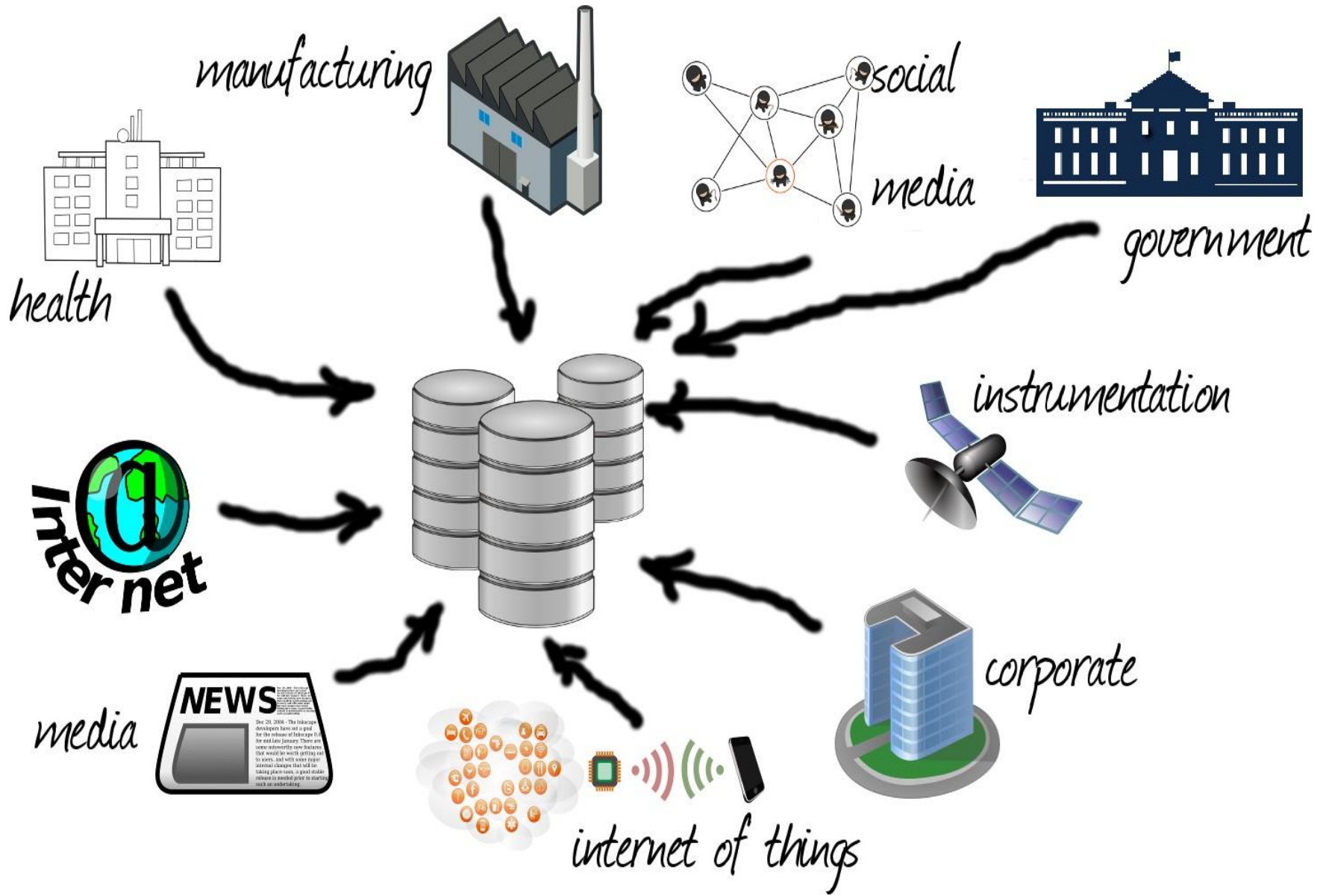
Data Science Components



Levels of Data Science



Data can be got from many sources



“Data is the New Oil”

Data

- ❖ Data refer to fact and Statistics collected together for reference and Analysis.
- ❖ Data can be collected/Stored
- ❖ Data can be Measured
- ❖ Data can be Analyzed
- ❖ Data can be visualized

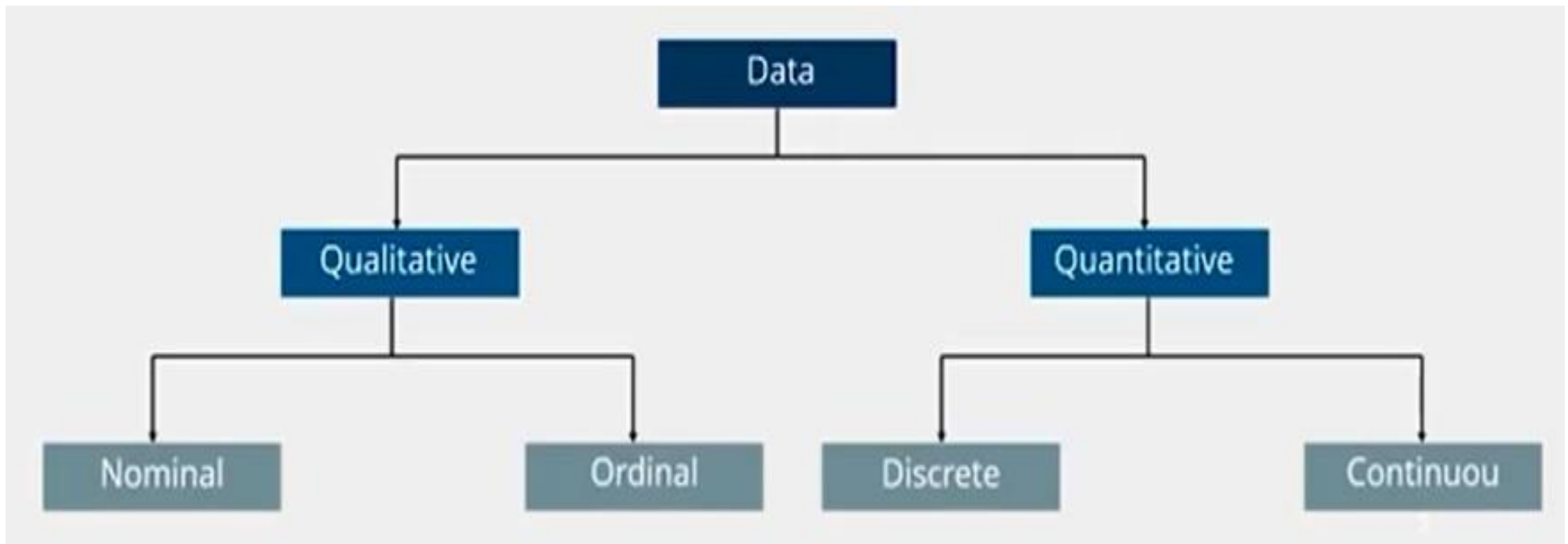
What is Data

Data refers to facts and Statistics collected together for reference or Analysis



Categories of Data

Data is categorized in to **Qualitative** and **Quantitative**



Qualitative Data

Qualitative data deals with characteristics and descriptors that can't be easily measured, but can be observed subjectively.

Nominal Data

Data with no inherent order or ranking such as gender or race, such kind of data is called Nominal data



Gender
Male
Female
Male
Male

Ordinal Data

Data with an ordered series, such as shown in the table, such kind of data is called Ordinal Data

Customer ID	Rating
001	Good
002	Average
003	Average
004	Bad

Quantitative Data

Quantitative data deals with numbers and things you can measure objectively

Discrete Data

Also known as categorical data, it can hold finite number of values

Example: Number of students in a class



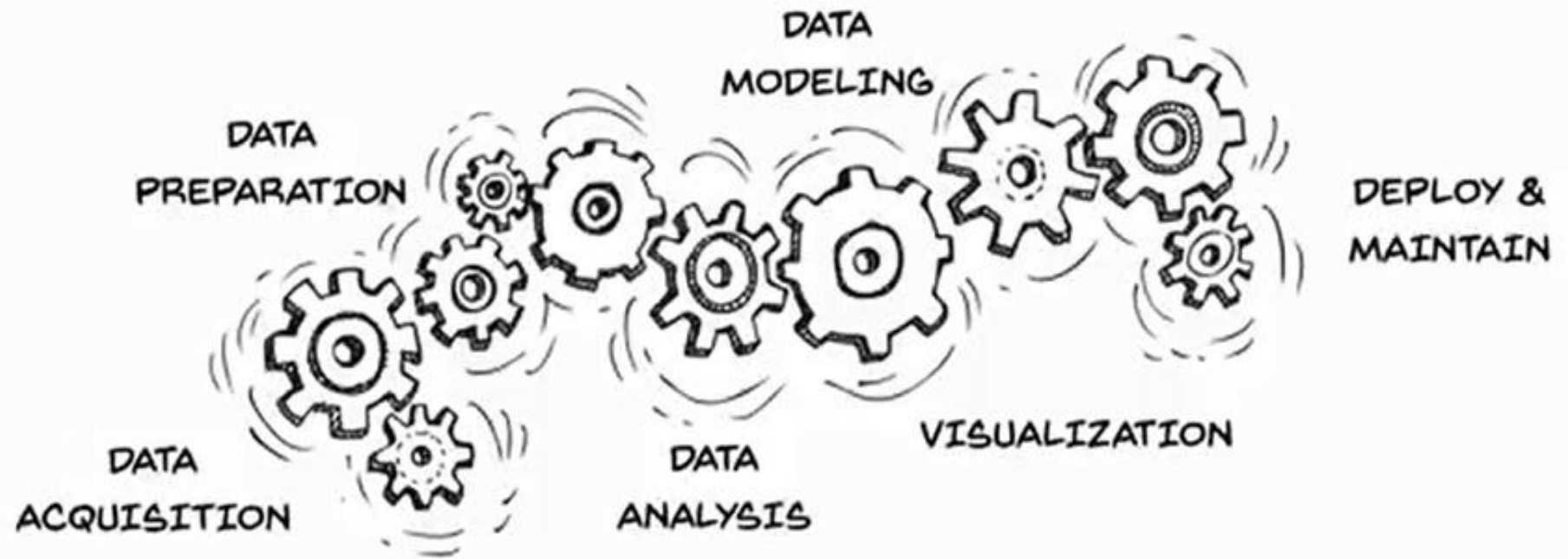
Continuous Data

Data that can hold infinite number of possible values

Example: Weight of a person



Data Science



Business Problem Understanding

①

BUSINESS PROBLEM

WHY?....WHY?....WHY?....



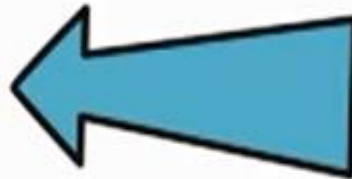
**ONE OF THE MANY TRAITS OF
A GOOD DATA SCIENTIST!**

Data Acquisition

②

DATA ACQUISITION

- WEB SERVERS
- LOGS
- DATABASES
- API'S
- ONLINE REPOSITORIES



Data Preparation



DATA PREPARATION

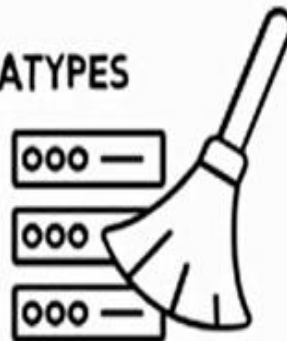
DATA CLEANING

TRANSFORMATION

INCONSISTENT DATATYPES

MISSPELLED ATTRIBUTES

MISSING AND DUPLICATE VALUES



Exploratory Data Analysis

④ EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) is an analysis approach that identifies general patterns in the data.

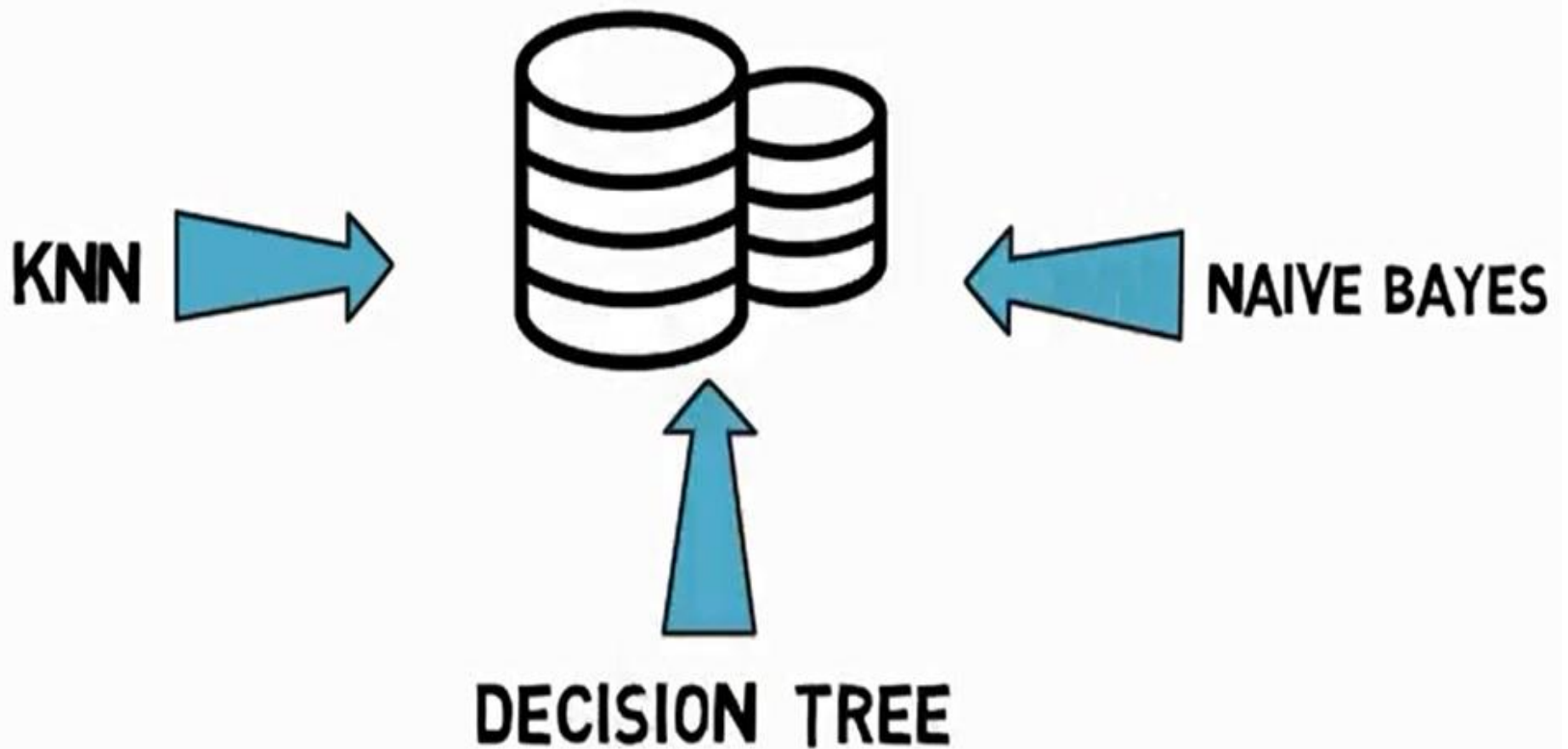


DEFINES AND REFINES
THE SELECTION OF FEATURE
VARIABLES THAT WILL BE USED
IN THE MODEL DEVELOPMENT

Data Modeling



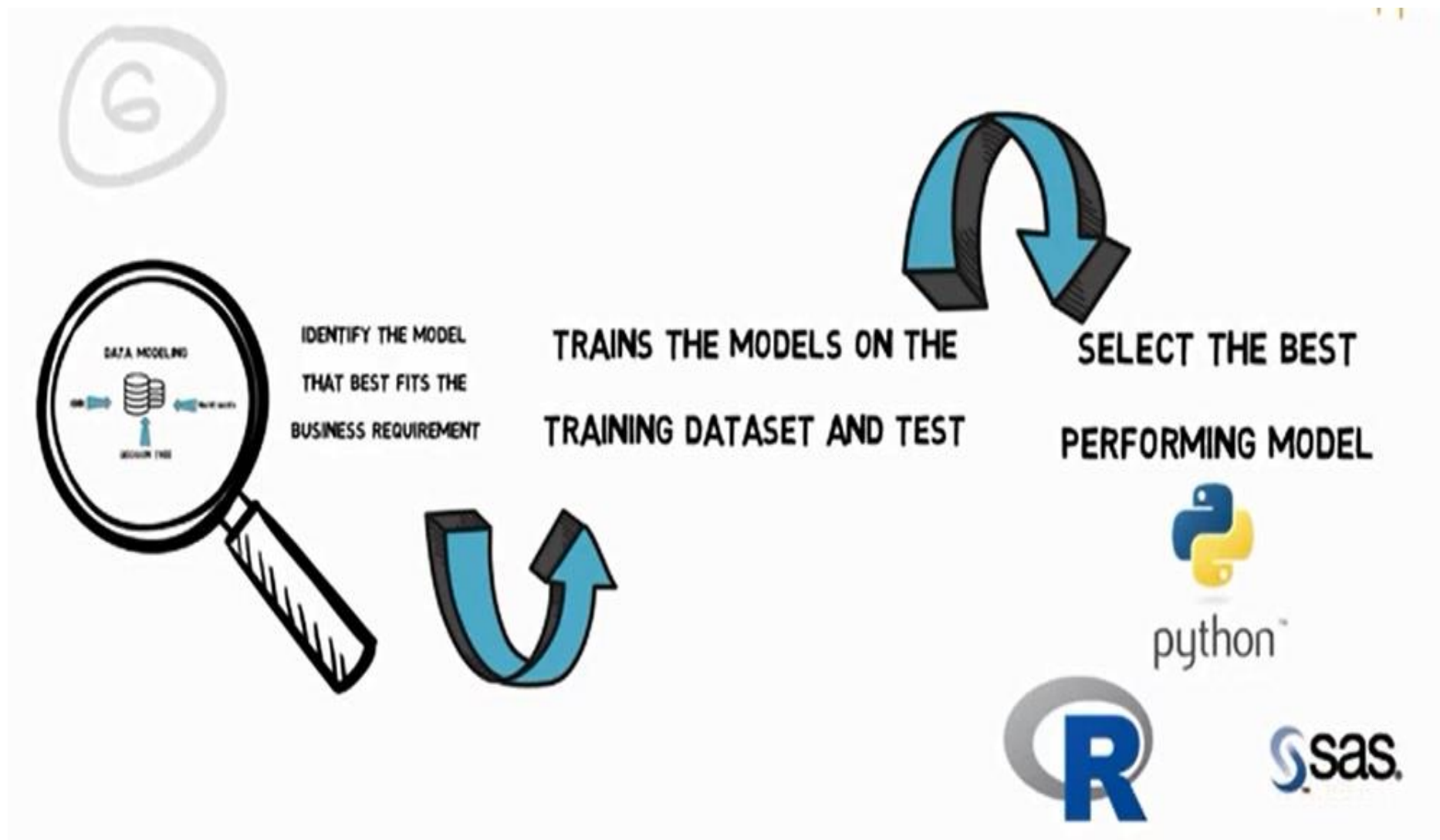
DATA MODELING



Difference Between Data Science and M L

Data science	Machine Learning
Data Science technique helps you to create insights from Data dealing with all Real-World Complexities.	Machine Learning method helps you to predict from historical data with the help of Mathematical models.
Data science can work with manual methods as well, though they are not very useful.	Machine learning algorithms hard to implement manually.
Data science is a complete process.	Machine Learning is a single step in the entire Data Science process.
Data Science is not a subset of Artificial Intelligence (AI).	Machine Learning is a subset of Artificial Intelligence (AI).

Data Modeling



Visualization and Communication

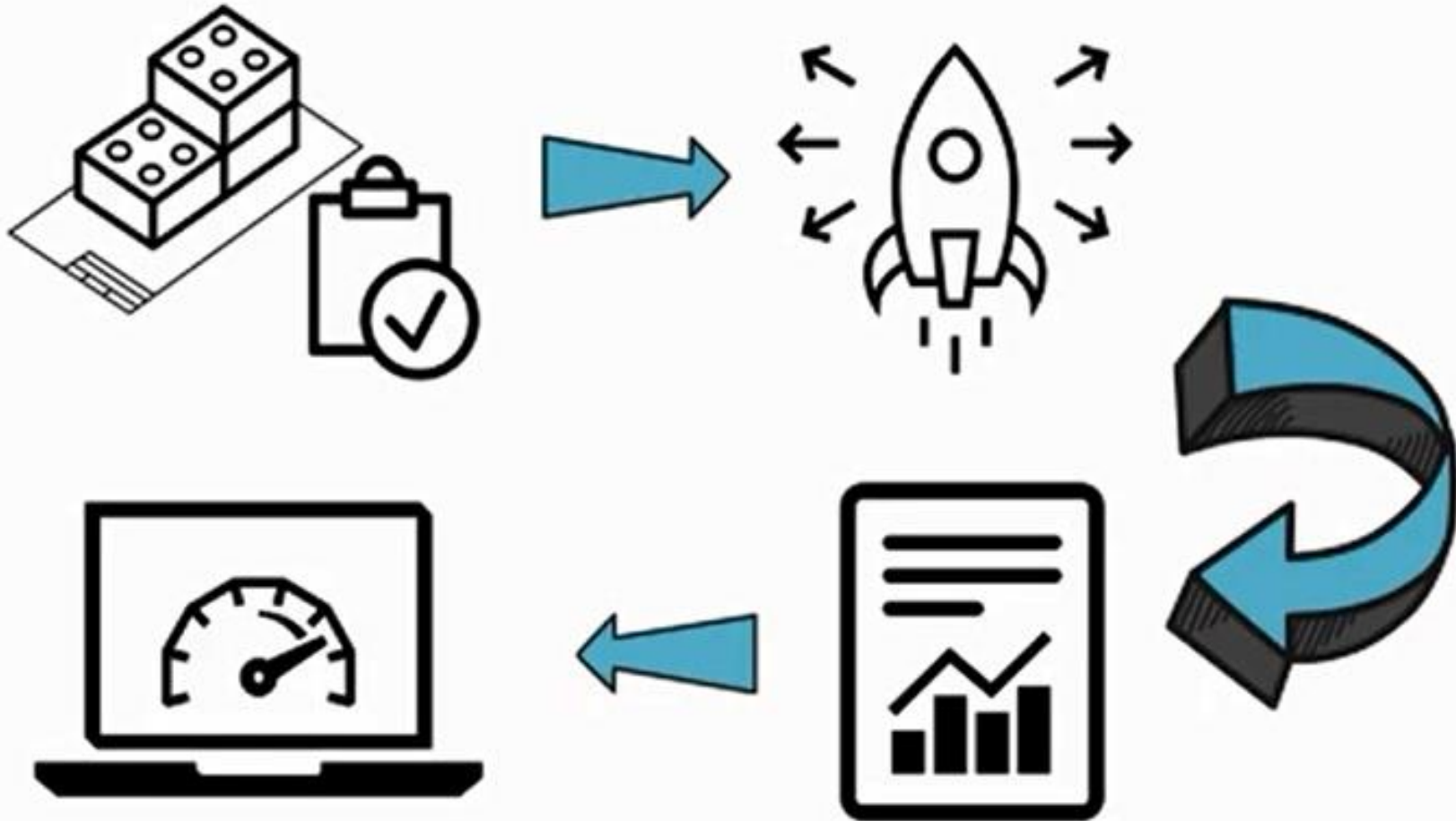
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VISUALIZATION AND COMMUNICATION

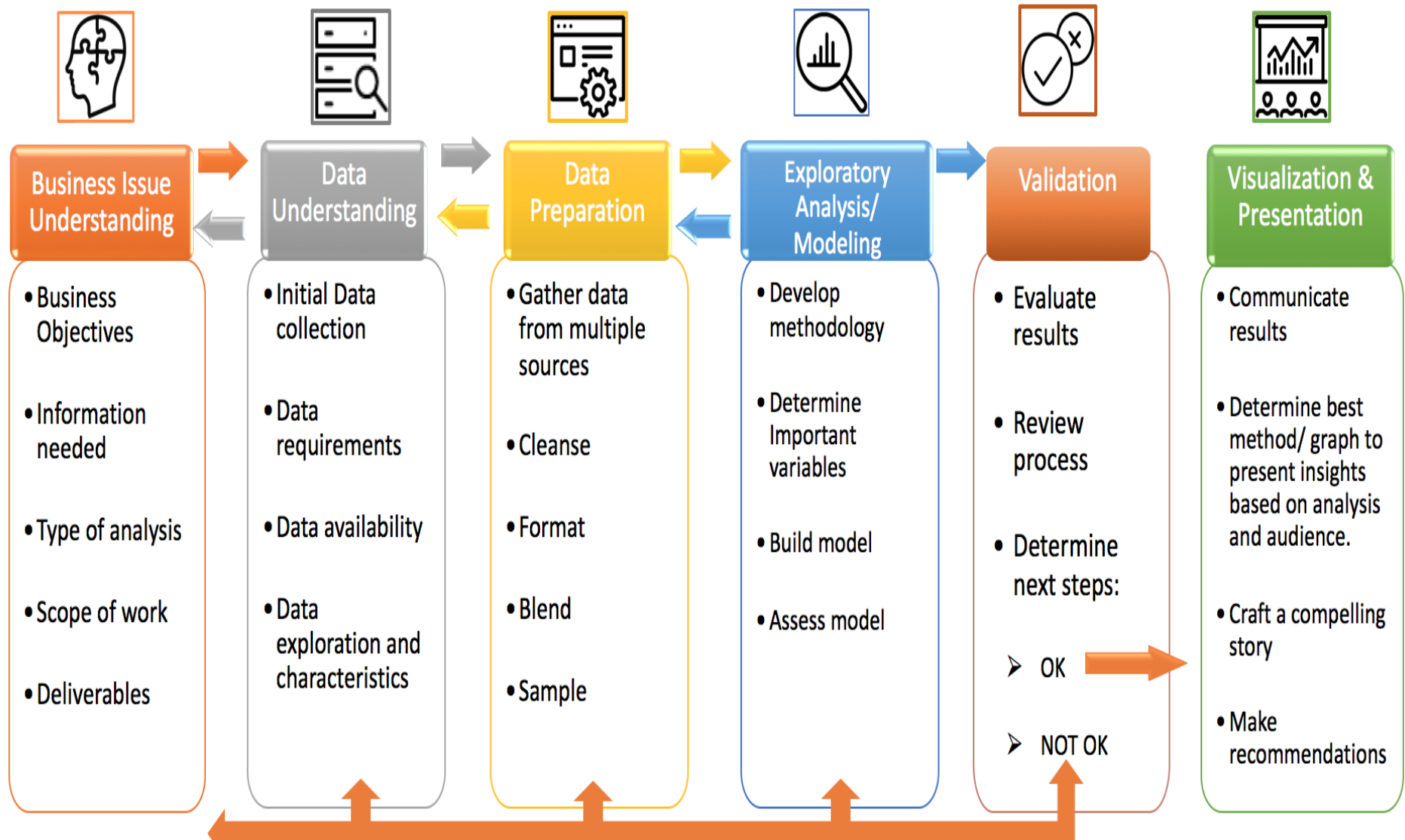


Deploys and Maintains

⑦ DEPLOYS AND MAINTAINS



Life Cycle of Data Science Projects



Parts of a Data Science Project

Collection: getting the data

Engineering: storage and computational resources

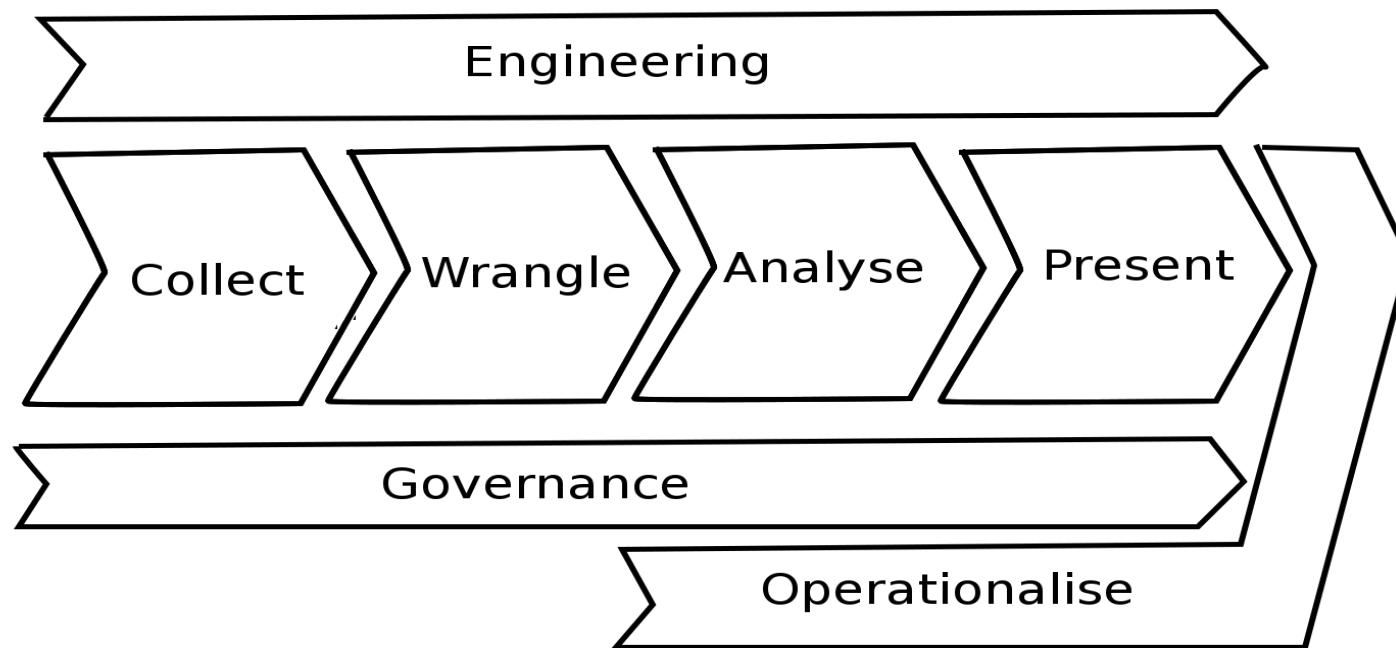
Governance: overall management of data

Wrangling: data preprocessing, cleaning

Analysis: discovery (learning, visualisation, *etc.*)

Presentation: arguing that results are significant and useful

Operationalisation: putting the results to work



Wrangling

If customer records are in 4 different databases in different formats; you want a single standardised set of customer names and addresses.

- ▶ Convert addresses in your customer database into geographic latitude and longitude.
- ▶ Convert free text dates to standard format, *e.g.* “next Tuesday”, “2nd January 15”, “January 3 next year”, “3rd Friday in the month”, “03/31/15”, “31/03/15”
- ▶ Recognize what values in your data are **“unknown”** or **“illegal”**

Tools for the Data Science-Common Softwares

Access: SQL, Hadoop, MS SQL Server, PIG, Spark

Data Analysis: R, Spark, Python and SAS

Wrangling: Common Scripting Languages (Python, Perl)

Visualization: Tableau, Matlab, Javascript+D3.js

Statistical Analysis: Weka, SAS, R

Multi-purpose: Python, R, SAS, KNIME, RapidMiner

Cloud-based: Azure ML (Microsoft), AWS ML (Amazon)

Python versus R

- ▶ Both are free
- ▶ R is better for Stand-alone analysis and exploration
- ▶ Python lets you integrate easier with other systems
- ▶ Python easier to learn and extend than R (better language)
- ▶ R has vectors and arrays as first class objects; similar to Matlab
- ▶ R currently less Scalable.

Fake News Detection

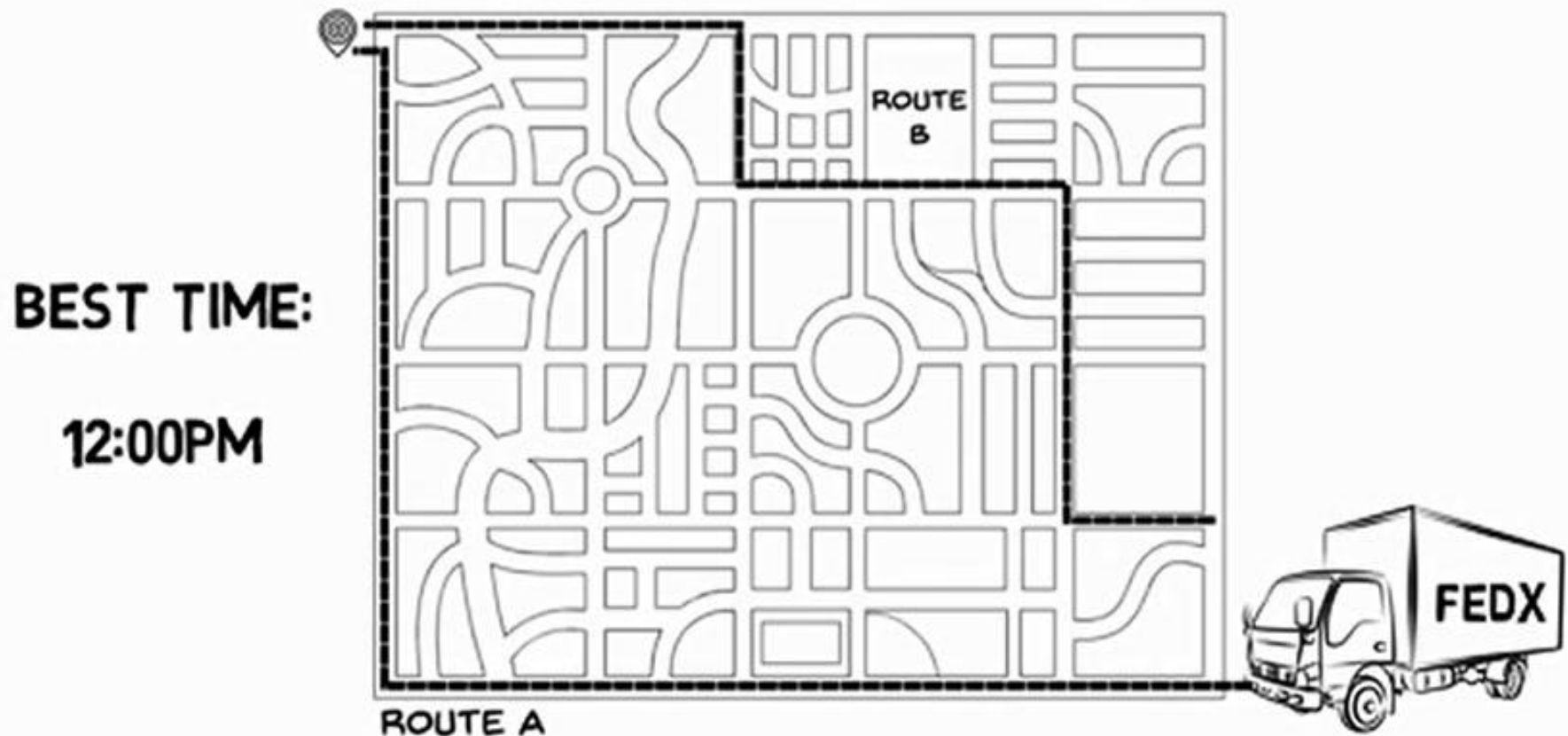
- ❖ With this data science project idea, you can use Python language to develop a specific model that can precisely detect whether the news is real journalism or false information.
- ❖ For this, 'Passive Aggressive Classifier' to classify the news into either a "Real" and "Fake" segmentations.
- ❖ The main idea of this Data Science project is to develop a real-time machine learning model that can correctly detect social media news authenticity.

Forest Fire Prediction

- ❖ One of the alarming & common disasters happening in today's world is forest fires. These disasters are highly damaging to the ecosystem.
- ❖ To deal with such a disaster, a lot of money on infrastructure & controlling and handling is required.
- ❖ We can build a Data Science project using 'k-means clustering'- it can identify any forest fires hotspots along with the severity of the fire at that particular spot.

Logistics

**LOGISTICS COMPANIES LIKE DHL, FEDEX HAVE
DISCOVERED THE BEST TIME AND ROUTES TO SHIP**



Airlines

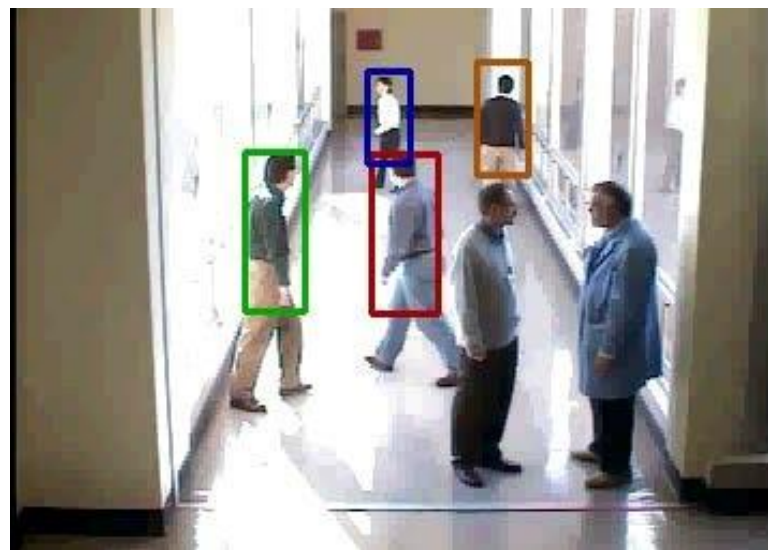
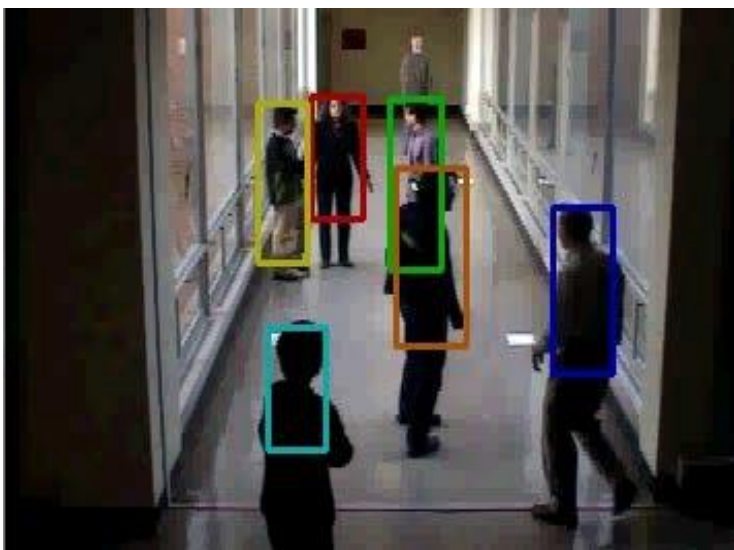
**AIRLINE COMPANIES CAN NOW EASILY PREDICT
FLIGHT DELAY AND NOTIFY THE PASSENGERS**



H487	CANCELLED
PP87	DELAYED

Revenue Management And Route Planning, Fuel Consumption And Optimization, Boarding And Checking Bags With Facial Recognition, Preparing Plane For Next Flight

Video Surveillance: Tracking



Waste Management and Healthcare Using AI

- ❖ The world produces over **2 billion tons of municipal solid waste** every year.
- ❖ The World Bank warns that global waste will increase up to 70 percent on current levels by 2050 unless **urgent actions are undertaken**.
- ❖ **Data science** can change the health care sector in so many ways.
- ❖ This, in turn, allows to **optimize the workforce and throughput, improves care recipients' satisfaction, and balances the supply**.



Expected outcome

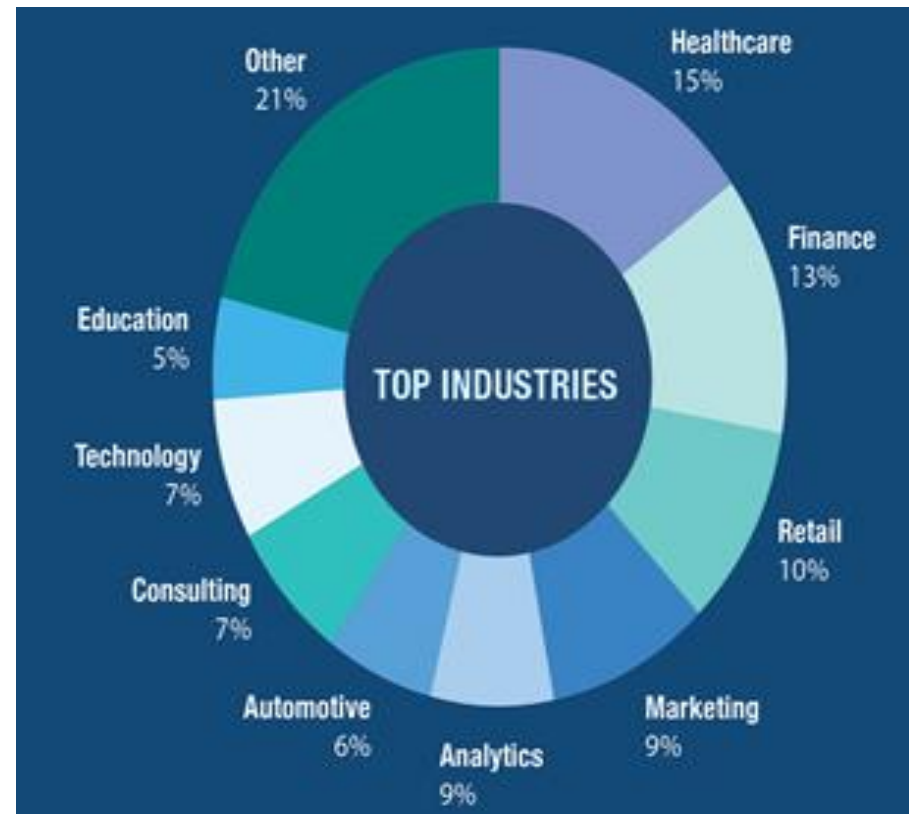
- Projects - Socially useful student projects, Research projects,
 - Start-up
 - Commercialize product
 - Publish- Quality papers, Patent
- Explore research areas in Waste management, Health Care using Data Analytics, Transportation, Logistics, Finance, Robotics etc.



Impact of AI on Industries

❖ AI Impact on Various Sectors

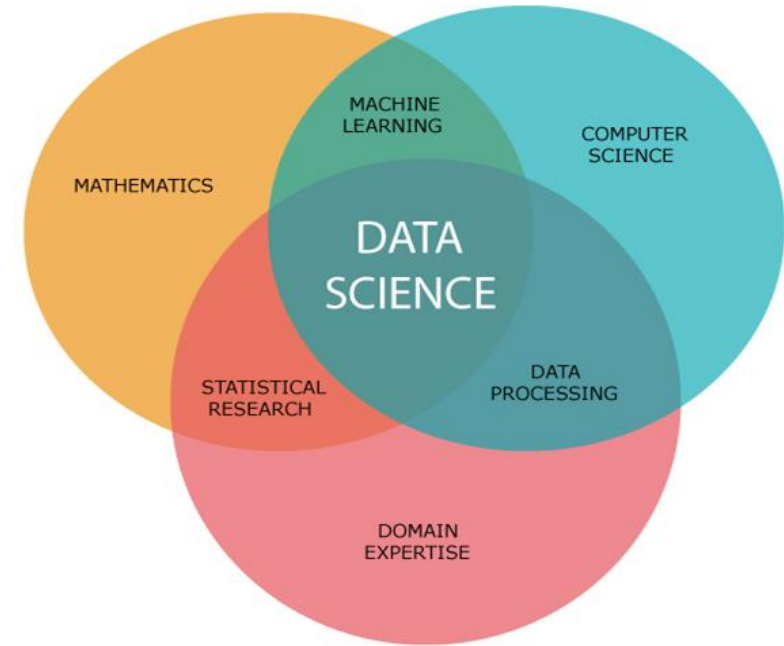
- ❑ Healthcare
- ❑ Automotive
- ❑ Financial Services
- ❑ Retail
- ❑ Technology
- ❑ Manufacturing
- ❑ Energy
- ❑ Transport and Logistics
- ❑ Robotics
- ❑ Analytics
- ❑ Education etc.



No sector or business is in any way immune from the impact of AI.

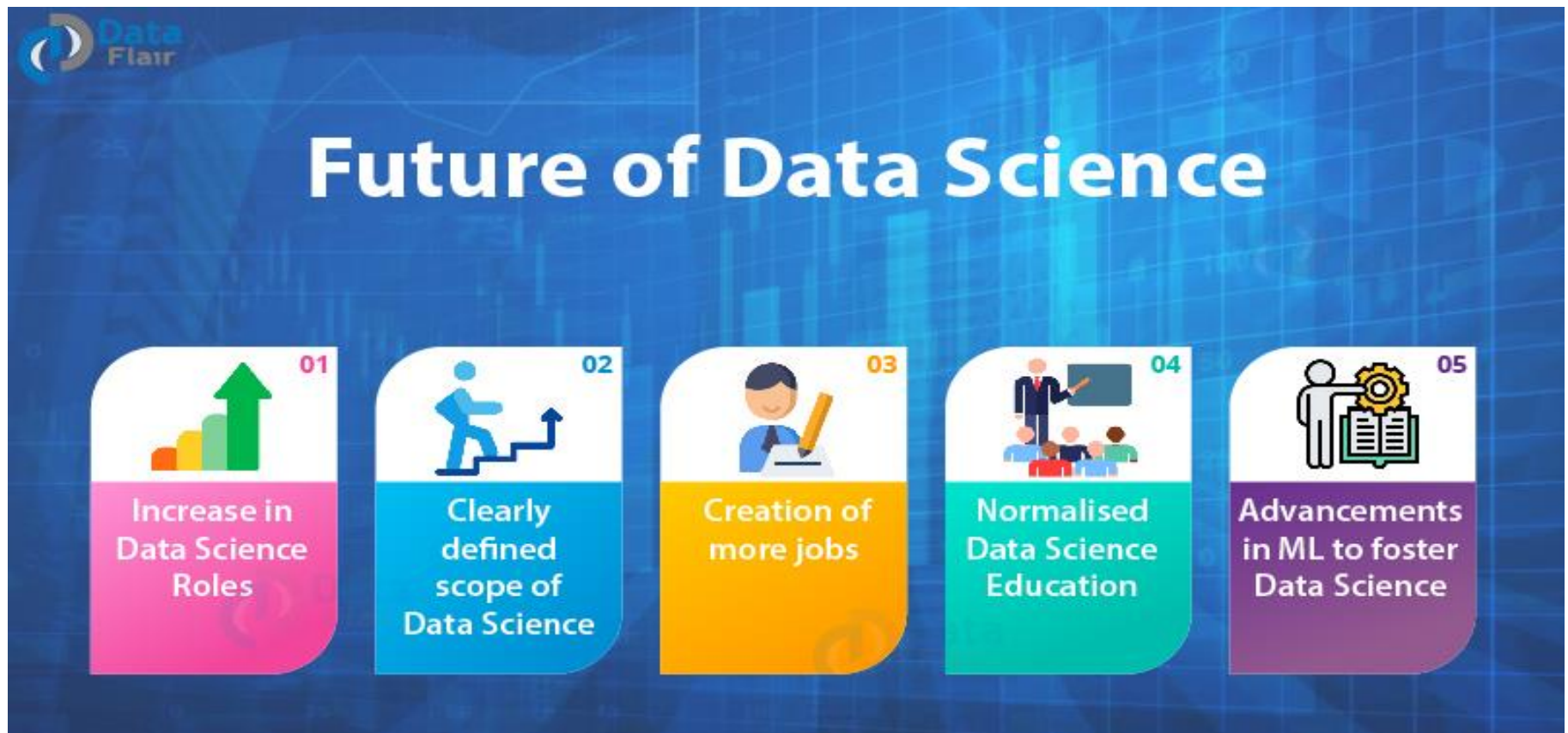
AI Engineer Skills

1. Analytical skills
2. Creativity
3. Data Driven Problem-solving skills
4. Critical thinking skills
5. Communication skills
6. Leadership skills
7. Self Learning
8. Mathematics and Algorithms
9. Cognitive Computing



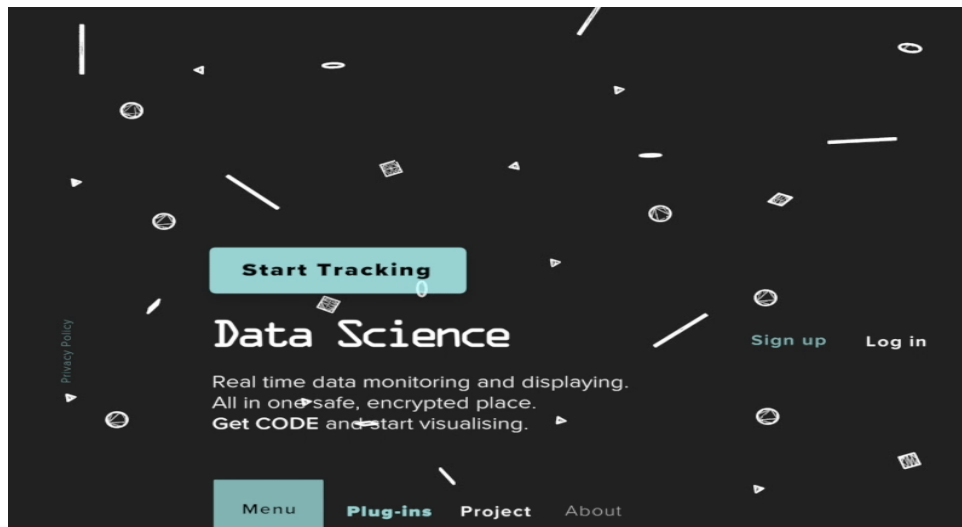
Market Trends and Projections

- **2.3 Millions of Jobs in Data Science in Future (Gartner Report).**
- Business Analyzing data are estimated to be around **\$430+ billions** by over 2025 over their rivals.
- **40% increase** in demands for **Data Scientist** and **Data Engineers** by 2026.



Research Opportunities

- Waste Management and Health Analytics using Data Science.
- Handling synchronized video analytics in a distributed cloud.
- Multiple and Heterogeneous information sources
- Predictive Business Analytics
- Computerizing front-end phases of the data life cycle
- Building domain-sensitive large scale frameworks
- Predictive Maintenance, Autonomous Car, & Surveillance etc.
- **Research Projects, Collaborative Projects** with Industry
- **Research Publications & Patents** etc.
- **Post covid-19 Situation**, there is **lot of scope for Start-ups** in **Data Science domain**.



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